

Routing

Course Code: CSC 3116

Course Title: Computer Networks



Dept. of Computer Science
Faculty of Science and Technology

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Lecture Outline



1. Introduction
2. Static routing
3. Dynamic routing
4. Routing information protocol (RIP)

Introduction



Routing is the act of forwarding **network packets** from a **source network** to a **destination networks** by a router based on its **routing table**.

CODE	NETWORK, MASK	AD/METRIC	NEXT HOP	INTERFACE
O	10.0.0.0 /8	110/20	200.1.1.1	S0
O	172.16.0.0 /16	110/15	200.1.1.1	S0
O	192.168.1.0 /24	110/20	200.2.2.2	S1
C	210.1.1.4 /30	0/0	Directly connected	E0

Fig. A simplified routing table

Introduction



❖ How is the routing table created?

- Static routing
 - If the **table is created manually** by inputting information for each destination network by a network engineer
 - Suitable for **very small network**
- Dynamic routing
 - If the **table is created and modified automatically** depending on the network condition by a routing protocol
 - Dynamic routing can be deployed on **small to large size network.**

Dynamic Routing

Features



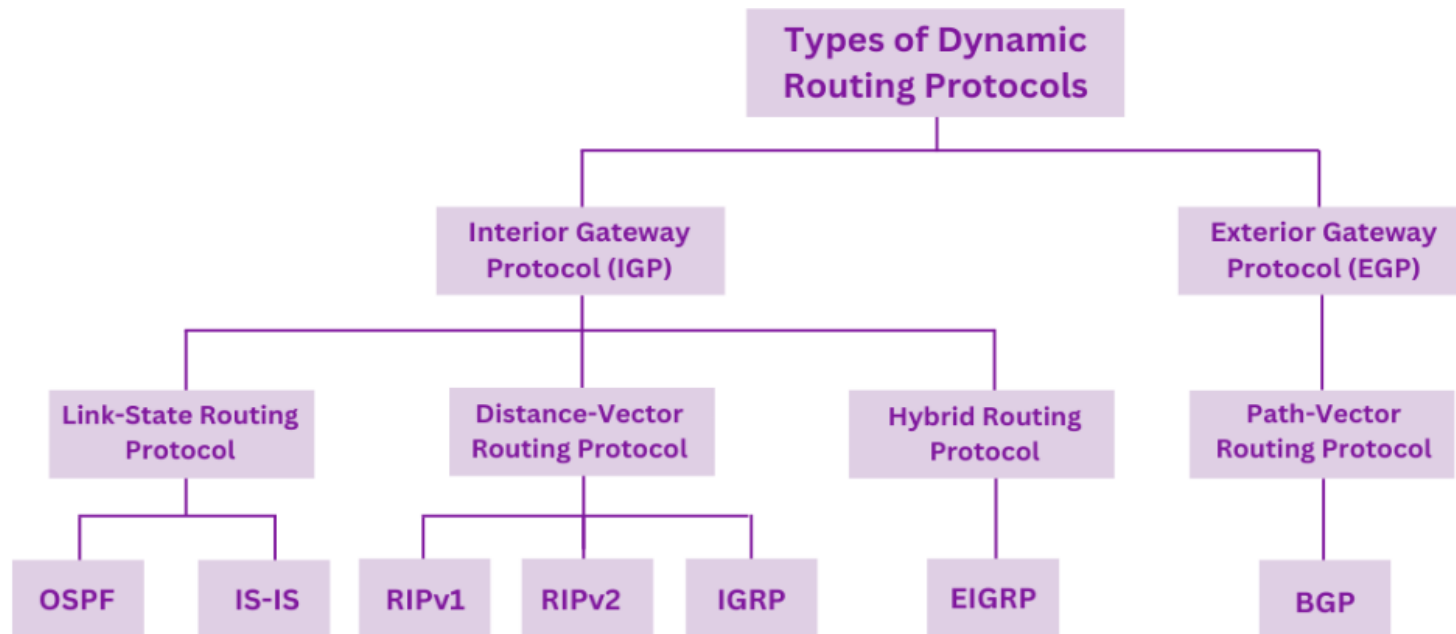
- Facilitates the **exchange of routing information** between routers
- Allow routers to **dynamically learn information** about remote networks and automatically add this information to their own routing tables
- Determines the best path to each network

Dynamic Routing....

Ways to classify routing protocols



- Interior Gateway Protocols (IGPs) or Exterior Gateway protocols (EGPs)
- Distance vector or Link state



Dynamic Routing....

IGP



□ IGP

- Routing inside an autonomous system (AS), where, AS is a collection of networks under **a common administrator**
- Example
 - Routing Information Protocol (RIP) version 1
 - Interior Gateway Routing Protocol (IGRP)
 - Enhanced IGRP (EIGRP)
 - Open Shortest Path First (OSPF)
 - Intermediate System (IS)-IS

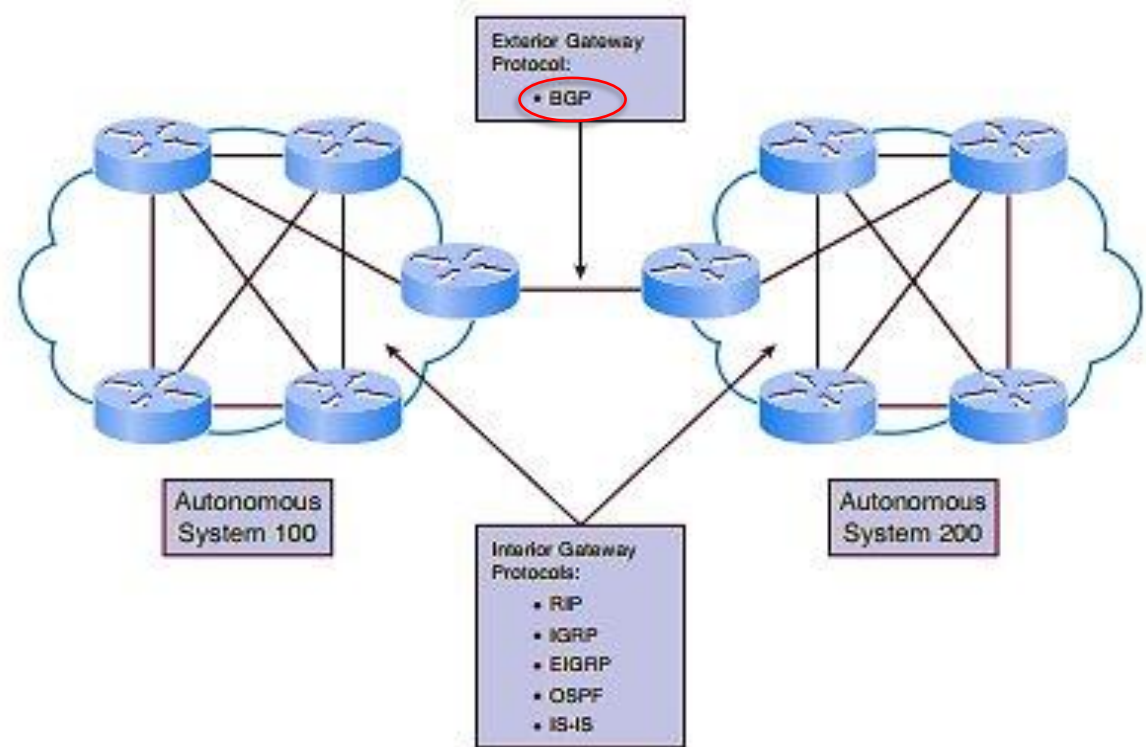
Dynamic Routing....

EGP



□ EGP

- Routing between autonomous systems (AS)
- Example
 - Border Gateway Protocol (BGP)

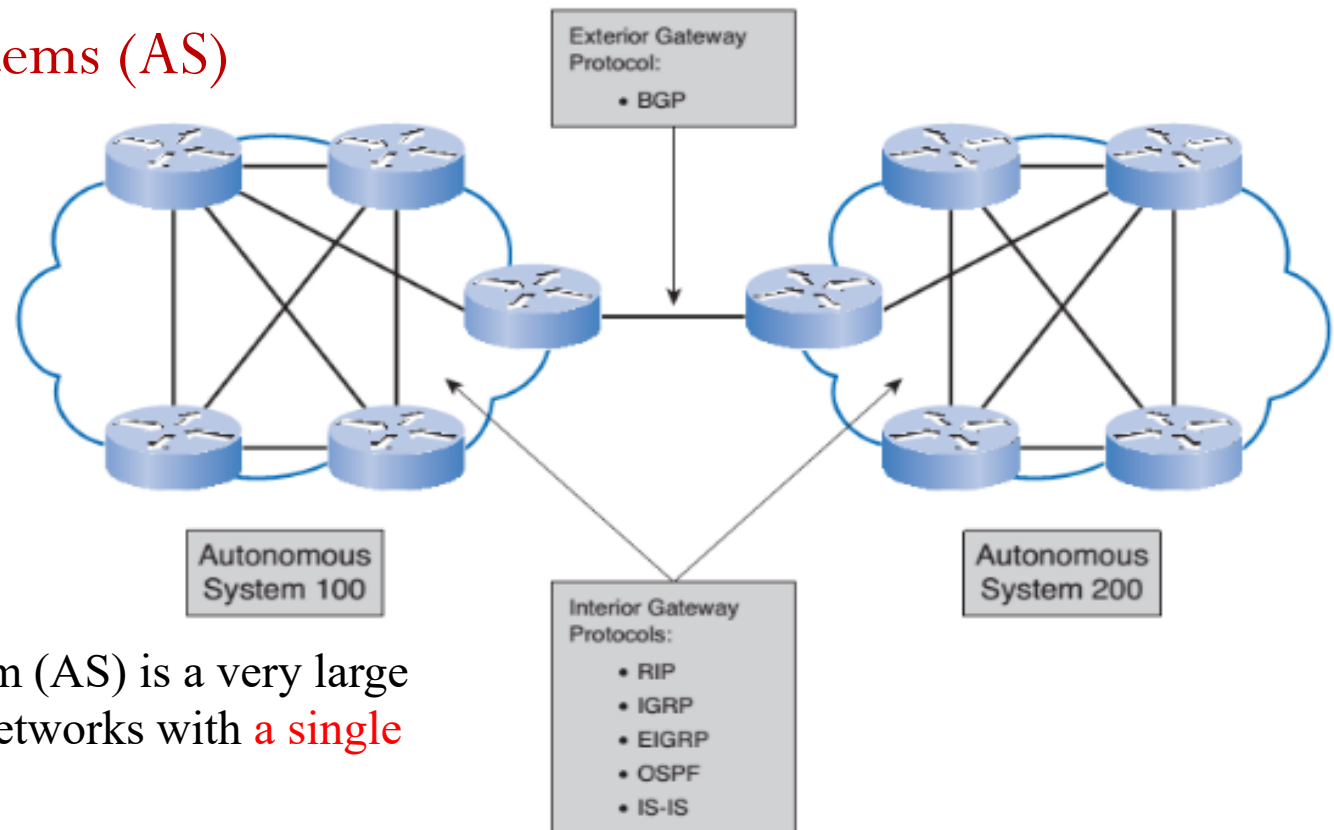


Dynamic Routing....

Classification



Autonomous systems (AS)



An autonomous system (AS) is a very large network or group of networks with **a single routing policy**.

Dynamic Routing....

Distance vector routing



- A router communicates with its neighbors only for populating its routing table
- *Distance vector* means that **routes are advertised** as *vectors* of distance and direction
- **Distance is defined** in terms of a metric such as **hop count**, and **direction is simply the next hop router or exit interface**.
- **Bellman-Ford algorithm** is used for the **best-path route determination**.
- Some distance vector protocols **periodically** send complete routing tables to all connected neighbors.
- In large networks, these **routing updates** can become enormous, **causing significant traffic** on the links.
- Example
 - RIP, IGRP, EIGRP (Hybrid)

Dynamic Routing....

Link state routing



- A router configured with a *link state routing protocol* can create a **“complete view,”** or topology, of the network by gathering information from all the other routers.
- A router communicates with all other routers of the network
- *Only send update (partial)* when there is any change in the network topology
- Example
OSPF

Dynamic Routing....

Administrative Distance



- The term *trustworthiness* is commonly used when defining administrative distance.
- Administrative distance (AD) defines the preference of a routing source.
- AD is an integer value from 0 to 255.
- The lower the value, the more preferred the route source.
- An AD of 0 is the most preferred.
- Only a directly connected network has an administrative distance of 0, which cannot be changed.

Route Source	AD
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

Dynamic Routing....

Administrative Distance



Routing Source	AD Value	Trust Level
Connected Interface	0	Most trusted
Static Route	1	Very trusted
EIGRP (internal)	90	Trusted
OSPF	110	Moderate trust
RIP	120	Least trusted
EIGRP (external)	170	Less trusted
Unknown/Invalid	255	Unusable (never trusted)

Route Source	AD
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

RIP

Features



- Distance vector routing protocol
- **Hop count** is used as the metric for path selection.
- If the **hop count** for a network is **greater than 15**, **RIP cannot supply a route to that network**.
- Routing updates are **broadcast or multicast every 30 seconds**, by default.
- If for some reason, **an update** for a **particular route** is not received **within a period of 180 seconds** then that specific route is declared **as invalid** and the router which identified that, informs all its neighbors about this invalid route.
- Has two versions: **RIPv1 & RIPv2**

RIP

RIP v1-vs-RIPv2



- **Uses Classless addressing**
RIPv1 uses **classful addressing**, **RIPv2 uses VLSM**
- **Multicasting vs Broadcasting**
RIP version 1 uses **broadcasting** (255.255.255.255) to send RIP messages to every neighbor. In this way, all the routers on the network receive the packets, as well as the hosts. **RIP version 2**, on the other hand, uses the all-router **multicast address** (224.0.0.9) to send the RIP messages only to RIP routers in the network.
- **Updates**
RIPv2 sends and receives version 2 updates only. RIPv1 sends version 1 updates and receives both 1 and 2, however, version 2 information is ignored.
- **Authentication**
RIPv2 ensures authentication, while RIPv1 does not



Multicast vs Broadcast



Feature	Broadcast	Multicast
Definition	Sends data to all devices in the network or subnet	Sends data to a group of selected devices only
Recipients	Everyone on the subnet receives it	Only subscribed members of the group receive it
Efficiency	Less efficient (wastes bandwidth)	More efficient , only reaches intended devices
IP Address Range	Uses IP: <code>255.255.255.255</code> (limited) or subnet broadcast	Uses IP range: <code>224.0.0.0</code> – <code>239.255.255.255</code>
Use Cases	ARP, DHCP Discovery	Streaming video, online gaming, webinars, stock feeds
Router Behavior	Routers block broadcasts by default across networks	Routers can forward multicast traffic if configured

RIP

Routing table update



❖ Cold Start



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

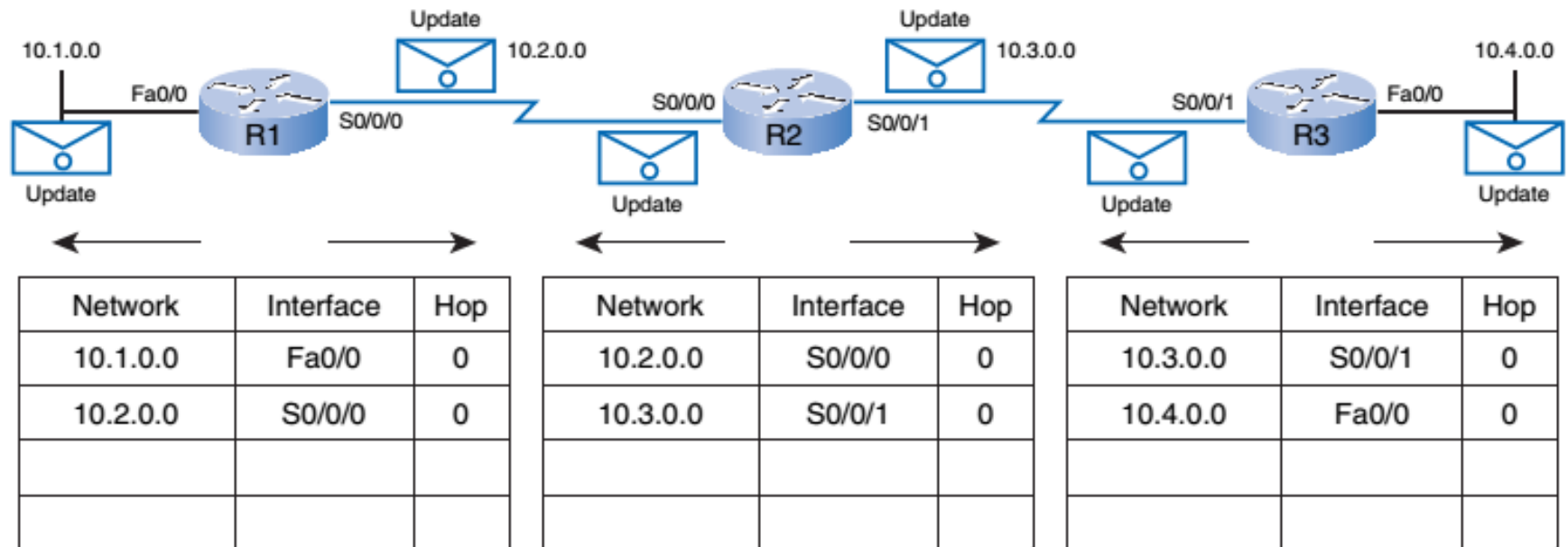
Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0

Topic Heading..

Routing table update

❖ Initial Exchange of Information



Topic Heading..

Routing table update



❖ Table Updating



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1



Topic Heading..

Routing table update

❖ Next Update



No New Information

Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

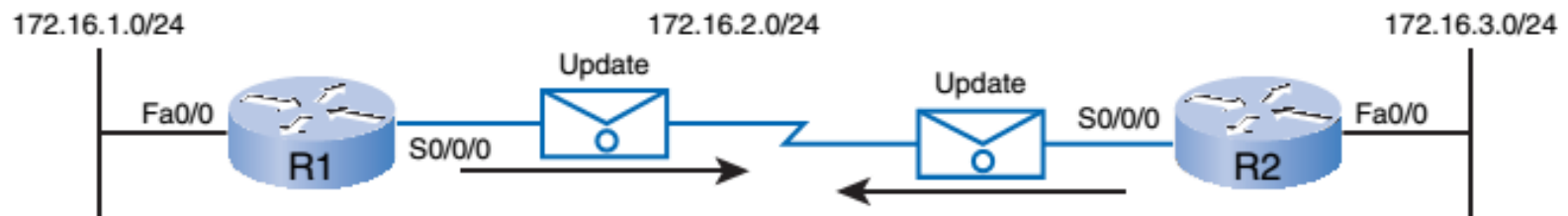
Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

↑
R1 and R3 now have complete routing tables.
↑

Topic Heading..

Another Example

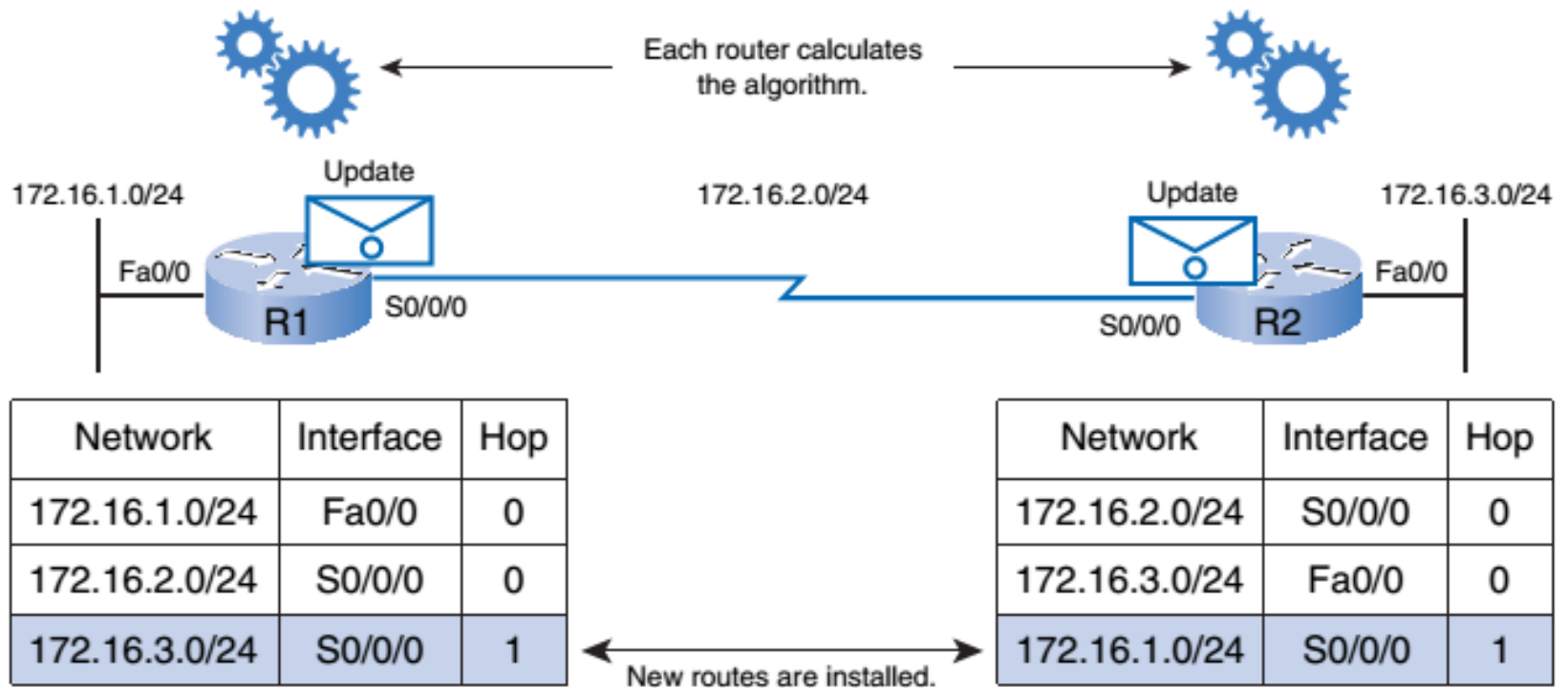


Network	Interface	Hop
172.16.1.0/24	Fa0/0	0
172.16.2.0/24	S0/0/0	0

Network	Interface	Hop
172.16.2.0/24	S0/0/0	0
172.16.3.0/24	Fa0/0	0

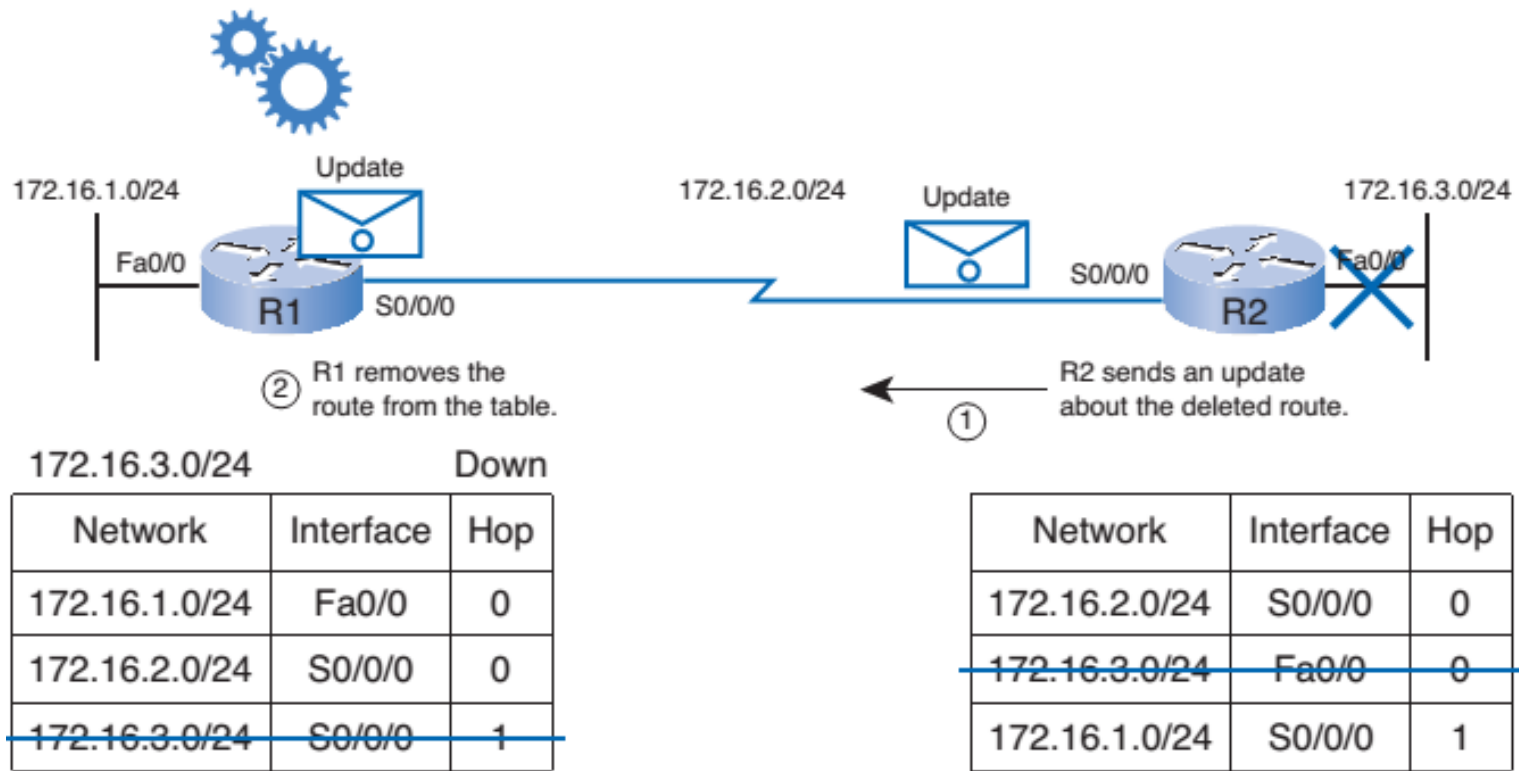
Topic Heading..

New route



Topic Heading..

Removing a route





References

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- [2] J. Macfarlane, *Network Routing Basics*, Wiley Publications. Inc., 2006, USA, pp. 70-104.



Books

- 1. Official Cert Guide CCNA 200-301 , vol. 1, *W. Odom*, Cisco Press, First Edition, 2019, USA.**
- 2. CCNA Routing and Switching, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.**